

CHEM 101 Topic 12: Introduction to Organic Chemistry — Answer Key

Section 1: What Makes Chemistry "Organic"?

Problem 1: CH₄ (Org), CaCO₃ (Inorg), C₃H₇OH (Org), KI (Inorg).

Problem 2: Carbon has 4 valence electrons and can bond to other carbons to form chains, rings, and branches.

Problem 3: Organic: Covalent, often liquid/gas. Inorganic Salt: Ionic, crystalline solid.

Section 2: Hydrocarbons

Problem 4: Alkane. (Fits C_nH_{2n+2}).

Problem 5: Propane.

Problem 6: C₄H₆. (Alkynes: C_nH_{2n-2}).

Section 3: Functional Groups

Problem 7: Aldehyde.

Problem 8: Carboxyl group. Example: Acetic acid.

Problem 9: Amine. Enables drug-receptor binding via polarity and H-bonding.

Section 4: Isomers and Biomolecules

Problem 10: Isomers.

Problem 11: Carbohydrates-C, Lipids-B, Proteins-D, Nucleic Acids-A.

Problem 12: Sequence -> Folding -> Shape -> Function.